THE EVOLUTION OF HEPATITIS C: FROM DISCOVERY TO CURE

Around 5000 years ago, the Sumerians first recorded an instance of epidemic jaundice, attributing it to demons (Campollo, Amaya, and McCormick, 2022). Though Hippocrates described this jaundice in "Epidemic" in the 5th century, its causes remained speculative throughout history. Unexplained jaundice cases continued to baffle doctors until 1969 when researchers identified a novel virus, 'non-A non-B hepatitis,' linked to blood transfusions. In 1977, pregnant Irish women were injected with anti-D immunoglobulin contaminated with this mysterious hepatitis form (Garvey et al., 2017). The Hepatitis C Virus (HCV) was identified and cloned in 1989. These women remained unaware of their Hepatitis C status until the mid-1990s (Garvey et al., 2017). Following the virus identification, epidemiological screenings estimated about 170 million people were infected globally (http://www.who.int). HCV was the leading cause of liver cirrhosis and cancer in the Western world. Although progress has been made in treating Hepatitis C, there are still 58 million chronic HCV cases, with approximately one million new infections annually.

The Causative Agent

The Hepatitis C Virus (HCV) is a small, enveloped virus from the Flaviviridae family, with a positive-sense, single-stranded RNA genome. It enters the body through contact with infected blood, targeting specific receptors on hepatocytes. Within the cell, the RNA genome is translated into a long non-functional polypeptide, which cleaves into structural (E1 and E2) and non-structural proteins (NS2, NS3, NS4A, NS4B, NS5A, and NS5B). Non-structural proteins, especially NS5A and NS5B, form replication complexes within membranous webs derived from the endoplasmic reticulum. The NS5B protein synthesizes a negative strand using the positive-sense RNA genome as a template, producing new positive-strand RNA genomes.

These genomes are packaged into virions near lipid droplets in the host cell. The mature virions are transported to the cell surface, released into the extracellular environment, and can infect new cells. During replication, the NS5B RNA-dependent RNA polymerase makes approximately 2.5×10^{-5} mutations per nucleotide per genome replication. This mutation rate, similar to RNA viruses like HIV and influenza, contributes to HCV's high genetic diversity (Ribeiro et al., 2012).

Traditional Therapies:

Before HCV identification, non-A non-B hepatitis was treated with Interferon therapy. Interferon monotherapy yielded sustained virological response (SVR) rates up to 20% after HCV identification (Manns et al., 2006). Interferon works nonspecifically against HCV by enhancing the body's natural immune response. A 12-month treatment regime yielded an SVR of 17%, significantly higher than controls (Poynard et al., 1996). However, interferon had severe side effects, with 25-50% of chronic HCV patients developing major depression (Oxenkrug et al., 2010). In the mid-1990s, treatment regimens began combining interferon with ribavirin. At 48 weeks, patients receiving the combination demonstrated an SVR of 38% compared to 19% for those treated with interferon alone (McHutchison et al., 1998). In the early 2000s, scientists developed pegylated interferon by attaching polyethylene glycol to the interferon molecule. This modification increased the half-life, reducing the frequency of doses and easing side effects. A study showed a combination regimen with peginterferon and ribavirin yielded an SVR of 54%, compared to 47% with the standard combination using interferon (Manns et al., 2001).

Generally, these combination regimens elicited SVR rates of about 50% (Pawlotsky, 2006). Thus, many with HCV were left without cures. These regimens were particularly

ineffective against HCV genotype 1 (Pawlotsky, 2006). Moreover, the lengthy treatments and severe side effects made compliance difficult.

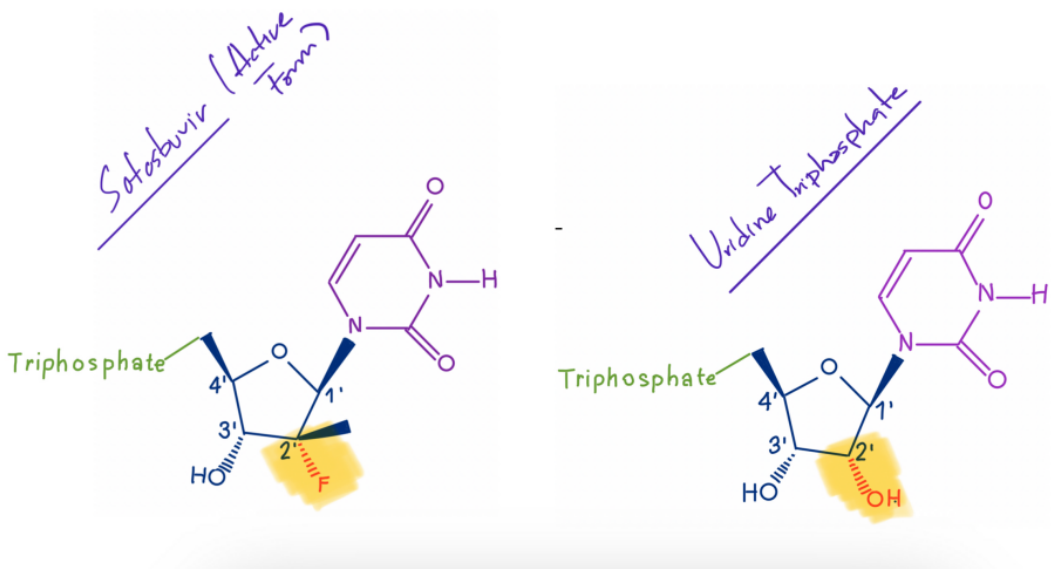
Early Advancements in DAA Development

In 2004, BLN 2061 (Ciluprevir) was orally administered to patients with HCV genotype 1, reducing plasma viral load by up to 3 logs in two days (Tsantrizos, 2004). Ciluprevir, targeting the NS3/4A protease, halts replication and new viral particle production by acting as a competitive inhibitor at the protease's active site. Despite its discontinuation due to toxicity, Ciluprevir demonstrated the potential of Direct-Acting Antiviral (DAA) agents, establishing proof-of-concept for HCV NS3 protease inhibitors (Tsantrizos, 2004). This led to developing first-generation DAAs, boceprevir and telaprevir, approved in 2011.

Telaprevir showed improved results over Ciluprevir, decreasing viral load by 1.3–5.3 log10 IU/mL within 15 days of monotherapy (Chatel-Chaix, Baril, and Lamarre, 2010). In 2011, a clinical trial confirmed the efficacy of Telaprevir combined with peginterferon and ribavirin versus standard treatments in HCV genotype 1 patients (Jacobson et al., 2011). While standard treatments had SVR rates of about 44%, the combination with Telaprevir improved SVR rates to 69-75%, completing treatments in 24 weeks instead of 48. However, Telaprevir's higher side effects led to more discontinuations, and drug-resistant mutations within the NS3/4A protease emerged within two weeks of monotherapy (Chatel-Chaix, Baril, and Lamarre, 2010). Additionally, the need for interferon-free treatments remained urgent.

In 2011, researchers identified the first potent NS5A protein inhibitor, crucial for RNA replication and viral particle assembly (Cordek et al., 2011). Daclatasvir inhibits NS5A protein dimerization and hyper-phosphorylation, essential for effective functioning of replication complexes (Gao et al., 2010). Around the same time, NS5B, the RNA-dependant polymerase

critical for RNA replication, also emerged as a critical target for DAAs (Powdrill et al., 2010). After catalysing the first phosphodiester bond de novo, NS5B adds additional nucleotides to the growing RNA chain to synthesise a complementary RNA strand based on the viral RNA template. During RNA synthesis, nucleotide inhibitors mimic the natural nucleotides and compete with them in order to be incorporated into the growing RNA chain. Particularly, Sofosbuvir contains a fluorine atom at the 2' position of the ribose sugar instead of the hydroxyl group that is present in natural uridine (Bhatia et al., 2014). When incorporated into the growing RNA chain instead of uridine, Sofosbuvir causes premature chain termination as the electronegative fluorine substitution lacks the nucleophilic character of the hydroxyl group that is necessary for the formation of phosphodiester bonds with adjacent nucleotides. Alternative to nucleoside inhibitors that target NS5B, non-nucleotide inhibitors bind to allosteric sites within the NS5B enzyme to induce conformational changes that effectively inhibit interactions between subdomains that are critical for RNA chain elongation (Wang et al., 2003).



In 2013, clinical trials confirmed Sofosbuvir's efficacy (Gane et al., 2013). Alone, Sofosbuvir achieved a 60% SVR rate in patients with HCV 2 and 3. Combined with ribavirin and peginterferon, it achieved a 100% SVR rate at 24 months after 12 weeks of treatment.

Notably, combination therapy without peginterferon also reached 100% SVR, marking the first effective interferon-free therapy. However, in HCV genotype 1 patients, the combination without peginterferon only achieved an 84% SVR rate. Therefore, researchers began combining different DAAs, especially with the development of NS5A inhibitors.

Resistance

Despite DAA progress, most treatment failures are due to resistance-associated substitutions (RAS). 88.6% of patients who failed DAA treatment had at least one RAS (Di Stefano et al., 2020). These mutations alter the HCV genome's amino acid sequence, affecting how viral proteins interact with antiviral agents, rendering them less effective. NS3/4 protease inhibitors are particularly susceptible to resistance (Dietz et al., 2017). Notably, Telaprevir monotherapy leads to RAS emergence within two weeks (Kieffer et al., 2007). The A156T mutation, substituting alanine with threonine at position 156 of the NS3/4A protease, significantly alters the binding pocket's shape and electrostatic properties, impairing binding efficacy (Welsch et al., 2008). This resistance mechanism is likewise to Sofosbuvir resistance development. Incoming nucleotides' hydroxyl groups form hydrogen bonds with serine at position 242, stabilizing nucleotides for RNA chain incorporation (Kulkarni et al., 2016). Sofosbuvir's fluorine component forms these hydrogen bonds less effectively. The S242T mutation substitutes serine for threonine, and the bulkier side chain's steric hindrance precludes hydrogen bonds with Sofosbuvir, preventing its incorporation into the RNA chain (Donaldson et al., 2015). Thus, Sofosbuvir can no longer induce chain termination, as the NS5B enzyme cannot stabilize it for effective incorporation into the RNA chain. Nevertheless, resistance against NS5B inhibitors like Sofosbuvir is less likely compared to NS3/4A protease inhibitors. While the S242T mutation impairs Sofosbuvir's binding efficacy, it also impairs replication

efficacy, making it rare (Svarovskaia et al., 2014). Moreover, multiple simultaneous mutations are often needed to inhibit Sofosbuvir's antiviral efficacy effectively. Comparatively, NS3/4A inhibitors, like Telaprevir, have a lower resistance barrier, as single mutations frequently confer high resistance levels without significantly impairing viral fitness (Welsch et al., 2008).

Modern Treatments

The advent of DAAs has been crucial in developing an HCV cure, but resistance challenges remain. Although Sofosbuvir is labeled a miracle cure, monotherapy only yielded a 60% SVR response rate. Among patients who previously failed Sofosbuvir-based therapies, adding Ledipasvir, an NS5A inhibitor, significantly improved the SVR response rate to about 98% (Wyles et al., 2015). These findings highlight the superiority of combining multiple DAAs over monotherapy. Studies suggest a lower RAS frequency when NS5B and NS5A inhibitors are combined (Pelosi et al., 2012). Such combination regimens target the virus at different points of its lifecycle, yielding a synergistic effect on inhibiting replication. Following the success of the ION trials, which established the efficacy of Sofosbuvir and Ledipasvir combination regimens (Nkuize et al., 2016), the ASTRAL trials investigated the success of Sofosbuvir and Velpatasvir (an NS5A inhibitor) combinations across all HCV genotypes (Hezodi et al., 2018). Results showed SVR response rates of up to 95-99%, achieving a cure for HCV effective across all genetic variations. In another study, Sofosbuvir and Velpatasvir, combined with the NS3/4A protease inhibitor Voxilaprevir, attained 100% SVR response rates in only 8 weeks among HCV genotype 1 patients (Gane et al., 2016).

Future Outlook

Despite advancements, a significant proportion of HCV-infected individuals remain undiagnosed. Globally, only 20% of people with chronic HCV have been diagnosed, and about 5% have initiated treatment (Martinello et al., 2020). Although we have the tools to treat HCV, eradicating it remains distant. Global HCV elimination likely requires a vaccine. However, HCV's inherent mutational capacity and high genetic variability complicate vaccine development. Progress has been made in developing neutralizing antibodies targeting conserved regions of the E1/E2 glycoprotein to elicit broad immune responses against multiple variants. For example, antibody 8D6 targets sites on the E2 glycoprotein critical for CD81 receptor interactions necessary for viral entry (Yi et al., 2020). Additionally, mutations at E2 glycoprotein allosteric sites, such as L442Q, can induce conformational changes, making targeted epitopes less accessible (Frumento et al., 2021). Moreover, adding a glycan at position N448 of the E2 glycoprotein can also mask targeted epitopes (Sevvana et al., 2021). Mutations within the E1 glycoprotein at position 348 can also propagate conformational changes that render neutralizing epitopes on E2 inaccessible to developed antibodies (Zhang et al., 2023). Nevertheless, research to understand how HCV evades the immune system and identify highly conserved neutralizing epitopes remains crucial for developing a vaccine.

Word Count: 1734

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Generative AI Acknowledgement

This report was written primarily through my own research and writing efforts. Generative AI (specifically, Notion AI) was used minimally to suggest improvements and enhance readability. The primary use of AI was to make direct suggestions for sentence structure and clarity. All content, including the historical context, chemical descriptions, and narrative, was researched and compiled independently by me. I cross-checked AI-suggested changes with authoritative sources to ensure accuracy and maintain the integrity of the information presented.

Additionally, the full report was modified using ChatGPT to minimize word count while ensuring all facts and points remain conserved and the writing stays true to my original work. The prompt used for this task was: "I will provide you with paragraphs I have written. You are to then reproduce these paragraphs. Your primary task is to alter the sequencing of words used and modify the combinations of sentences that are written to eliminate redundancy in order to rewrite the paragraph as efficiently as possible, minimizing word count. However, keep my writing as similar as possible, ensuring that all facts and points presented remain conserved and the writing remains my original work. Your job is to only reduce the word count by making minor modifications and also to improve spelling and grammar."

Here is the link to the input provided to chatGPT:

https://docs.google.com/document/d/1ujKOLGPJR8ZwlWEq45UNq6xvlxpIK28-2qzpYUrghzE/edit?usp=sharing